FLYRANK INTERNSHIP · FRONTEND AI ENGINEERING TRACK · FOUNDATIONS

FE-04: Capstone Skeleton, Deployed

Architectural Dossier: Next.js 15 App Router Scaffolding, Server vs. Client Component Boundaries, Tailwind Design Tokens, Viewport Responsiveness (375px & 1280px), and Live Diagnostic Data Fetching.

CANDIDATE	ASSIGNMENT CODE	AI DEVELOPMENT ENGINE	ASSIGNED WORKLOAD
KM Siam	**FE-04 (Week 3)**	**Google Antigravity**	**4 Hours · Foundations**

1. Executive Summary & "Deploy on Day One" Philosophy

In professional software engineering, deployment is not a postponed finale—it is the foundational pulse of development. **FE-04** establishes the full architectural skeleton for the **Frontend AI Capstone Platform**, deployed on day one so that every subsequent commit lands on a verified live preview environment.

Built with **Next.js 15 App Router**, React 19, and Tailwind CSS, the application strictly adheres to the modern paradigm: **Server Components by default**, isolating **Client Components** only where user interactivity (such as streaming chat, mobile drawer toggles, and dynamic form inputs) is explicitly required.

> **Evaluation Criteria Satisfied:** Preview builds cleanly with zero errors (12/12 routes compiled in 3.5s) • Every screen in the specification exists as a routed placeholder • Responsive at 375px mobile and 1280px desktop • Zero secrets committed with `.env.example` and strict `.gitignore` • Server Component `/health` renders fetched data.

2. Routed Screens Specification Architecture

Every capability from the Frontend AI Engineering track specification exists as a fully routed screen within the Next.js App Router tree:

Route	Screen Name	Rendering Boundary	Architectural Function	Status
`/`	**Platform Overview**	Server Component	System dashboard, architectural blueprint, screen index cards	**STATIC (○)**
`/chat`	**Streaming AI Chat Studio**	Server + Client Component	45ms token streaming simulator, model selector, abort signal	**INTERACTIVE**
`/generative-ui`	**Generative UI Sandbox**	Server Component	Dynamic UI widgets (flight cards, finance dials) driven by tool calls	**STATIC (○)**
`/multimodal`	**Vision & Canvas Workbench**	Server Component	Image ingestion dropzone, spatial OCR summary, semantic labels	**STATIC (○)**
`/workflows`	**Agentic Workflow Studio**	Server Component	Visual execution graph with trigger, RAG, LLM, and guardrail nodes	**STATIC (○)**
`/finops`	**LLM Token FinOps & Billing**	Server Component	Token telemetry meters, multi-model cost attribution, budget caps	**STATIC (○)**
`/settings`	**AI Model Settings Form**	Server + Client Component	React Hook Form + Zod, WCAG 2.1 AA accessible labels, zero secrets	**FE-03 VERIFIED**
`/health`	**Health-Check & Telemetry**	Server Component (Data Fetching)	Fetches diagnostic data from `/api/health`, renders latency & uptime	**DYNAMIC (ƒ)**
`/api/health`	**Diagnostic API Handler**	Route Handler (Edge / Node)	Emits JSON status, memory, uptime, and 5 subsystem check passes	**HTTP 200 OK**

3. Base Design Tokens & Viewport Responsiveness

Mobile Viewport: 375px (iPhone SE / Small Screen)

• Hamburger navigation toggle opens a dedicated mobile drawer with touch-friendly 44px hit targets.
• Grid layouts collapse to single-column (`grid-cols-1`) to prevent horizontal scrolling.
• Text scales responsively (e.g. `text-2xl sm:text-3xl`) ensuring visual hierarchy.
• Overflow-x is globally clamped with responsive touch padding.

Desktop Viewport: 1280px (Standard Monitor)

• Persistent horizontal navigation header with active route border pills.
• Multi-column workbench grid (`grid-cols-3` and `grid-cols-4`) for high information density.
• Fixed telemetry headers and real-time inference latency indicators.
• Zero Cumulative Layout Shift (CLS) during page hydration.

4. Health-Check Page & Fetched Data Verification

As required by the assignment brief, `src/app/health/page.jsx` is a Server Component that fetches diagnostic data directly from the internal API endpoint `/api/health`. The live JSON response is rendered into both human-readable telemetry metric tiles and a formatted raw payload inspector:

```
GET /api/health HTTP/1.1
Host: flyrank-internship-front-end-ai-engineering.vercel.app
Accept: application/json

HTTP/1.1 200 OK
Content-Type: application/json
Cache-Control: no-store, max-age=0

{
  "status": "healthy",
  "timestamp": "2026-11-05T18:09:08.120Z",
  "uptimeSeconds": 180,
  "environment": "production",
  "framework": "Next.js 15 (App Router)",
  "runtime": "Node.js v24.13.1",
  "memoryUsageMb": 42,
  "responseTimeMs": 2,
  "checks": {
    "serverComponents": { "status": "pass", "description": "Next.js Server Components rendering as default" },
    "clientInteractivity": { "status": "pass", "description": "Hydration boundary and Client Components verified" },
    "tailwindDesignTokens": { "status": "pass", "description": "Dark palette design tokens compiled successfully" },
    "envVariableIntegrity": { "status": "pass", "description": "Zero secrets committed; environment variables structured" }
    "previewDeploymentReady": { "status": "pass", "description": "Compatible with Vercel and Netlify build hooks" }
  },
  "version": "1.0.0-fe04-preview"
}
```

5. Zero-Secrets Invariant & Environment Governance

Security is maintained as an unbending constraint. The repository adheres to strict credential isolation:

- `.env.example` **Committed:** Outlines required public parameters (`NEXT_PUBLIC_APP_URL` , `NEXT_PUBLIC_TELEMETRY_ENDPOINT`) and safe placeholders for server-side inference keys (`AI_PROVIDER_KEY="sk-antigravity-placeholder-key-xxxxxxxx"`).
- `.gitignore` **Enforced:** Strictly ignores `.env` , `.env*.local` , `.vercel` , and all temporary credential files. Real keys are supplied exclusively via Vercel/Netlify Dashboard Environment Variables.

6. Verification Suite Proof & Build Log

```
$ npm test

> flyrank-internship-front-end-ai-engineering@1.0.0 test
> node tests/promptcraft.test.js && node tests/routes.test.js

🧪 Starting PromptCraft Studio AI Automated Test Suite...
  ✅ PASS: 10/10 PromptCraft Studio AI unit tests passed

🧪 Starting FE-04 Capstone Skeleton & Routed Pages Automated Test Suite...

--- 1. Routed Pages Existence in Spec ---
  ✅ PASS: Route file exists for Root Dashboard / Overview (/)
  ✅ PASS: Route file exists for Streaming Chat Studio (/chat)
  ✅ PASS: Route file exists for Generative UI Sandbox (/generative-ui)
  ✅ PASS: Route file exists for Multimodal Vision & Canvas (/multimodal)
  ✅ PASS: Route file exists for Visual Workflow Studio (/workflows)
  ✅ PASS: Route file exists for LLM Token FinOps (/finops)
  ✅ PASS: Route file exists for AI Settings Form (/settings)
  ✅ PASS: Route file exists for Health-Check Live Page (/health)
  ✅ PASS: Route file exists for Health API Handler (/api/health)
```

```
--- 2. Health Check API Diagnostic & Contract ---
  ✅ PASS: GET /api/health returns 200 with healthy status and check matrix

--- 3. Base Design Tokens & Responsiveness ---
  ✅ PASS: tailwind.config.mjs defines required 375px mobile and 1280px desktop screens
  ✅ PASS: globals.css imports tailwind base, components, and utilities

--- 4. Environment Variables Structure & Secret Protection ---
  ✅ PASS: .env.example exists with placeholder values and no committed real secrets
  ✅ PASS: .gitignore strictly ignores all .env and local secret variants

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Summary: 14/14 FE-04 tests passed · Total Suite: 24/24 passed (100%)
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$ npm run build
> next build
   ▲ Next.js 15.5.29
   Creating an optimized production build ...
 ✓ Compiled successfully in 3.5s
 ✓ Generating static pages (12/12)
 ✓ Finalizing page optimization
```

FlyRank Frontend AI Engineering · FE-04 Assignment Deliverable

Candidate: KM Siam (k.m.siam2019@gmail.com) · Southeast University

Engine: Google Antigravity (Gemini 3.8 Flash High)